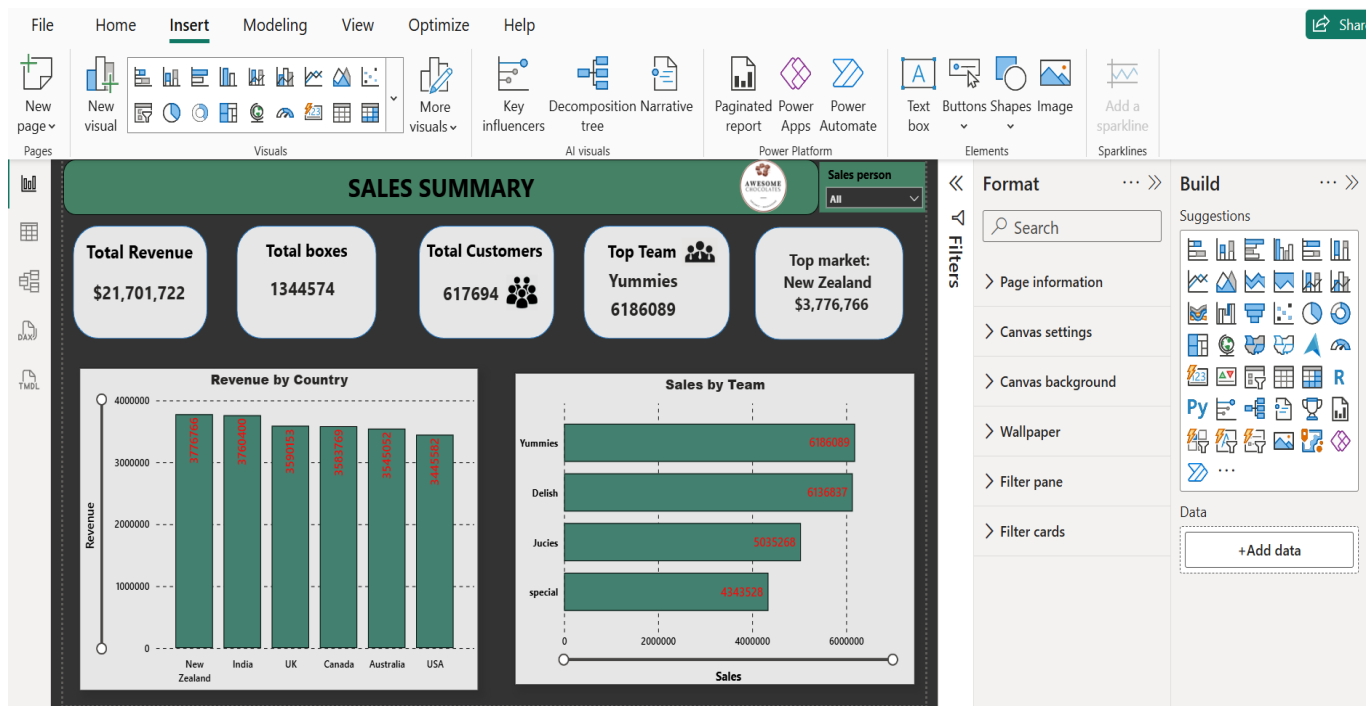


AWESOME CHOCOLATES

Sales Performance Dashboard

A Power BI Portfolio Project



Link to the project: https://drive.google.com/file/d/1f5zo7iuvzBkuL9KJijRcrGYnONqfAG1k/view?usp=drive_link

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Hi, I'm Madalitso

Welcome to my project! I'm Madalitso Lungu, an Economics and Statistics graduate based in Lusaka, Zambia, and this is one of the projects I've built while growing my skills in data analytics. I'm currently building out my portfolio as I look for opportunities in data analytics and economics roles, and this dashboard is one of the pieces I have worked on.

I decided to build this portfolio because I wanted more than certificates and course completions to show for the time I've spent learning Power BI. I wanted a real, working example of how I think through a business problem, clean messy data, model it properly and turn it into something a manager could actually use to make decisions. This project uses a fictional but realistic dataset from “Awesome Chocolates,” a chocolate distributor selling across six countries, and it gave me the chance to practice the full journey: from a raw, imperfect spreadsheet all the way to an interactive, multi-page dashboard.

My longer-term goal is to move into financial research analysis, and building strong data storytelling habits now asking the right business questions, being honest about what the data can and can't tell you is groundwork I want to get right early. I hope this walkthrough gives you a clear sense of not just what I built, but how and why I built it the way I did.

The Problem This Dashboard Solves

Awesome Chocolates sells across six markets New Zealand, India, the UK, Canada, Australia, and the USA through four sales teams and dozens of individual salespeople. Like a lot of growing sales operations, the raw transactional data existed, but nobody could easily answer the questions that actually matter day to day:

- Where is our revenue actually coming from, and which markets deserve more attention?
- Who are our strongest performers, and is that because of skill, territory, or something else?
- Are we growing, shrinking, or flat and what should we expect next month?

This dashboard was built to turn scattered transaction-level data into a tool a sales director or operations manager could open on a Monday morning and immediately understand where the business stands without needing to run a single manual report.

Tools and Techniques I Used

I wanted this project to demonstrate more than dragging fields onto a canvas, so I leaned into proper data modelling and DAX rather than relying only on default visuals.

Tools

- Power BI Desktop: data modelling, DAX measures, report design
- Power Query: data cleaning and transformation (details below)
- Star schema data modelling: one fact table (Sales) connected to Products, People, and Locations dimension tables

Techniques

- DAX time intelligence (DATESINPERIOD, EOMONTH) for month-over-month comparisons
- RANKX and TOPN for dynamic “top performer” logic that updates as filters change
- VAR-based measures for readable, debug gable DAX rather than deeply nested formulas
- Conditional formatting and measure-driven font colours (e.g. green/red trend indicators)
- Native Power BI forecasting (exponential smoothing with 95% confidence intervals)
- Synced slicers for a consistent, cross-page filtering experience

DATASET

Our data consist of a location, people, product and sales table.

									
Product	Category	Size	Geo	Region	Sales person	Team	Picture		
Milk Bars	Bars	LARGE	India	APAC	Barr Faughny	Yummies	https://files.chandoo.org/pbix/img/men-1.jpg		
50% Dark Bites	Bites	LARGE	USA	Americas	Dennison Crosswaite	Yummies	https://files.chandoo.org/pbix/img/men-2.jpg		
Almond Choco	Bars	LARGE	Canada	Americas	Gunar Cockshoot	Yummies	https://files.chandoo.org/pbix/img/men-3.jpg		
Raspberry Choco	Bars	LARGE	New Zealand	APAC	Wilone O'Kielt	Delish	https://files.chandoo.org/pbix/img/men-4.jpg		
Mint Chip Choco	Bars	LARGE	Australia	APAC	Gigi Bohling	Delish	https://files.chandoo.org/pbix/img/men-5.jpg		
Eclairs	Bites	LARGE	UK	Europe	Curtice Advani	Delish	https://files.chandoo.org/pbix/img/men-6.jpg		
Drinking Coco	Other	LARGE			Kaine Padly	Delish	https://files.chandoo.org/pbix/img/men-7.jpg		
99% Dark & Pure	Bars	LARGE			Ches Bonnell		https://files.chandoo.org/pbix/img/men-8.jpg		
Orange Choco	Bars	LARGE			Andria Kimpton	Jucies	https://files.chandoo.org/pbix/img/men-9.jpg		
Spicy Special Slims	Bites	LARGE			Brien Boise	Jucies	https://files.chandoo.org/pbix/img/men-10.jpg		
After Nines	Bites	LARGE			Husein Augar	Yummies	https://files.chandoo.org/pbix/img/women-11.jpg		
Fruit & Nut Bars	Bars	LARGE			Karlen McCaffrey	Yummies	https://files.chandoo.org/pbix/img/women-12.jpg		
85% Dark Bars	Bars	SMALL			Jan Morforth	Delish	https://files.chandoo.org/pbix/img/women-13.jpg		
White Choc	Other	SMALL			Dotty Strutley	Delish	https://files.chandoo.org/pbix/img/women-14.jpg		
Baker's Choco Chips	Bars	SMALL			Kelci Walkden	Yummies	https://files.chandoo.org/pbix/img/women-15.jpg		
Organic Choco Syrup	Other	SMALL			Marney O'Brien	Jucies	https://files.chandoo.org/pbix/img/women-16.jpg		
Caramel Stuffed Bars	Bars	SMALL			Rafaelita Blaksland		https://files.chandoo.org/pbix/img/women-17.jpg		
Manuka Honey Choco	Other	SMALL			Madelene Upcott	Jucies	https://files.chandoo.org/pbix/img/women-18.jpg		
70% Dark Bites	Bites	SMALL			Beverie Moffet	Jucies	https://files.chandoo.org/pbix/img/women-19.jpg		
Smooth Silky Salty	Bars	SMALL			Oby Sorrel	Jucies	https://files.chandoo.org/pbix/img/women-20.jpg		
Choco Coated Almond	Bites	SMALL			Mallorie Waber		https://files.chandoo.org/pbix/img/women-21.jpg		
Peanut Butter Cubes	Bites	SMALL			Jehu Rudeforth		https://files.chandoo.org/pbix/img/women-22.jpg		
					Van Tuxwell	Yummies	https://files.chandoo.org/pbix/img/men-11.jpg		
					Roddy Speechley	Delish	https://files.chandoo.org/pbix/img/men-12.jpg		
					Camilla Castle		https://files.chandoo.org/pbix/img/women-23.jpg		

Sales Person	Geography	Product	Date	Amount	Customers	Boxes	Day of Week	Day Number
Barr Faughny	New Zealand	Raspberry Choco	01 January 2021	8414	276	495	Friday	5
Barr Faughny	New Zealand	Baker's Choco Chips	01 January 2021	259	32	22	Friday	5
Curtice Advani	New Zealand	Milk Bars	01 January 2021	12894	115	478	Friday	5
Wilone O'Kielt	New Zealand	Organic Choco Syrup	01 January 2021	6377	184	246	Friday	5
Dennison Crosswaite	New Zealand	85% Dark Bars	01 January 2021	2506	99	148	Friday	5
Jehu Rudeforth	New Zealand	Peanut Butter Cubes	01 January 2021	3409	93	171	Friday	5
Roddy Speechley	New Zealand	Caramel Stuffed Bars	08 January 2021	2716	299	105	Friday	5
Rafaelita Blaksland	New Zealand	Peanut Butter Cubes	08 January 2021	2373	2	104	Friday	5
Gigi Bohling	New Zealand	50% Dark Bites	08 January 2021	6797	114	486	Friday	5
Jehu Rudeforth	New Zealand	85% Dark Bars	08 January 2021	6048	56	319	Friday	5
Rafaelita Blaksland	New Zealand	Raspberry Choco	15 January 2021	4949	244	248	Friday	5
Curtice Advani	New Zealand	99% Dark & Pure	15 January 2021	308	91	19	Friday	5
Dotty Strutley	New Zealand	Caramel Stuffed Bars	15 January 2021	6895	318	300	Friday	5
Dennison Crosswaite	New Zealand	Mint Chip Choco	15 January 2021	3430	120	202	Friday	5
Karlen McCaffrey	New Zealand	Orange Choco	22 January 2021	4312	181	616	Friday	5
Camilla Castle	New Zealand	85% Dark Bars	22 January 2021	1358	68	68	Friday	5
Marney O'Brien	New Zealand	50% Dark Bites	22 January 2021	3038	280	254	Friday	5
Ches Bonnell	New Zealand	Baker's Choco Chips	22 January 2021	5971	59	498	Friday	5
Husein Augar	New Zealand	Almond Choco	29 January 2021	10941	185	730	Friday	5
Camilla Castle	New Zealand	Almond Choco	29 January 2021	9142	1	436	Friday	5
Wilone O'Kielt	New Zealand	Mint Chip Choco	12 February 2021	6503	299	407	Friday	5

able: sales (3,791 rows)

Update available (click to download)

Link for full set dataset

https://drive.google.com/file/d/1f5zo7iuvzBkuL9KJijRcrGYnONqfAG1k/view?usp=drive_link

Data Cleaning Procedure

Before any of the analysis could be trusted, the data needed cleaning. This step mattered more than I expected a data quality issue I found partway through actually changed how one of my charts looked.

- **Whitespace trimming:** several text fields including Geography, Sales Person, and Product had inconsistent leading spaces. Power BI treats these as entirely different values which meant some transactions weren't matching correctly to the Locations dimension table and were showing up as a “(Blank)” category on my Revenue by Country chart. I used Power Query's Transform → Format → Trim function across the affected text columns to fix this.
- **Correct data typing:** confirmed Date, Amount, Boxes, and Customers were typed correctly (date and numeric, not text) so that aggregation and time-based DAX functions would work as expected.
- **Relationship validation:** checked that Sales connected cleanly to Products, People, and Locations with the correct one-to-many cardinality, since a mismatched relationship can silently produce wrong totals.

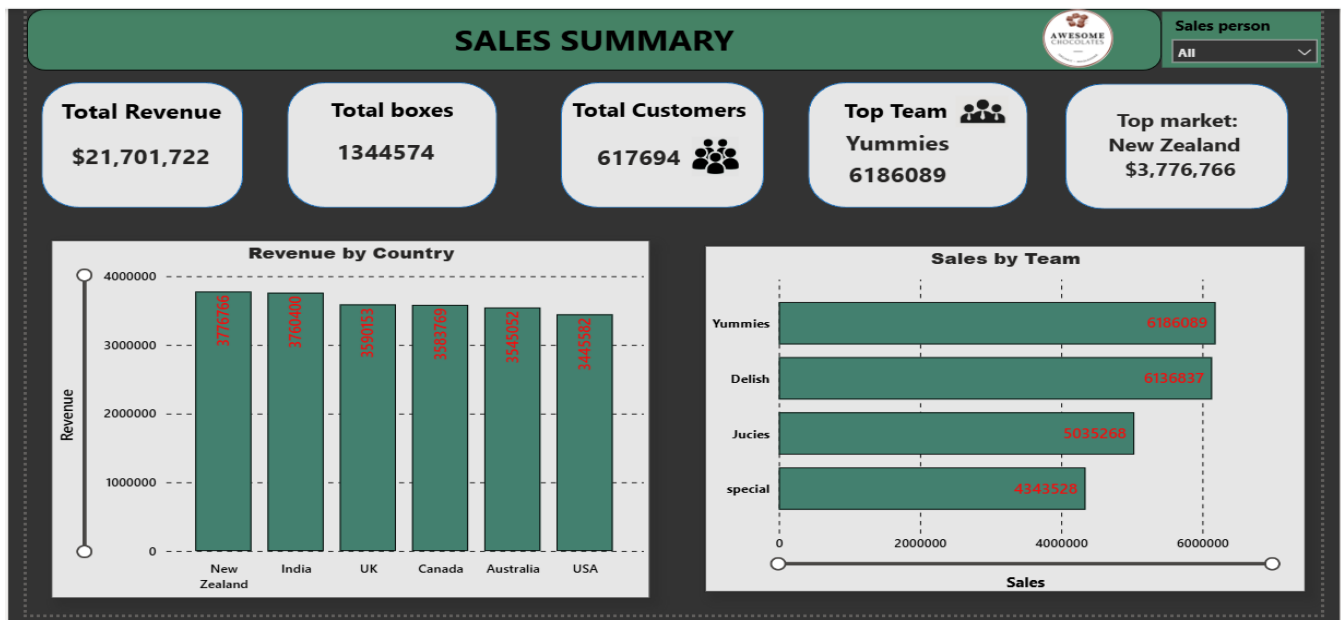
The whitespace issue is a good example of why I think data cleaning deserves real attention rather than being treated as a formality it was an invisible problem until I noticed an unexplained “(Blank)” bar, and tracing it back taught me to check for it as a first step on any new dataset going forward.

The Dashboard, Page by Page

The dashboard is built as a three-page story: Overview → Sales Performance → Trend & Forecast. A synced Sales Person filter lets you follow one person's story across all three pages without losing your place.

Page 1: Sales Summary

This page answers the question: “At a glance, how is the business doing, and where is that coming from?” It opens with five KPI cards Total Revenue, Total Boxes, Total Customers, Top Team and Top Market followed by a revenue by country chart and a sales-by-team breakdown.



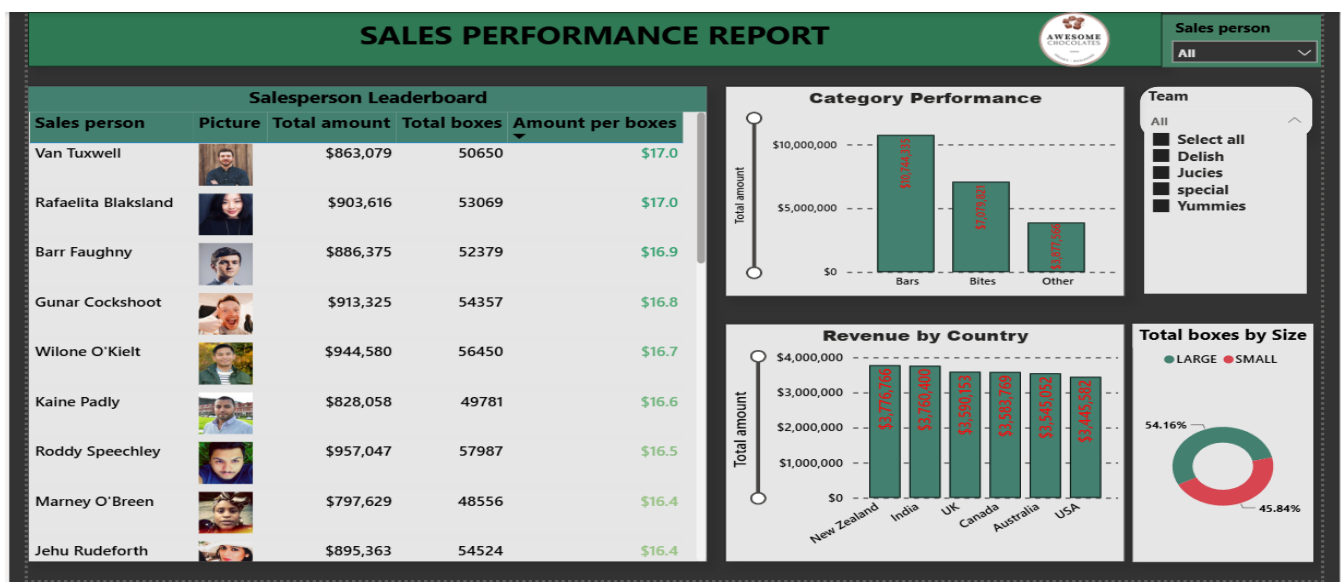
Page 1: Sales Summary overview

Scenario

A regional manager opens the dashboard Monday morning before a leadership call. In ten seconds, they can see total revenue sits at \$21.7M, New Zealand is the top market at \$3.78M, and the Yummies team is leading overall enough context to speak confidently without pulling a manual report.

Page 2: Sales Performance Report

This page answers: “Who is actually driving our numbers, and how efficiently?” Rather than just ranking people by total revenue, I included an Amount per Box measure a simple efficiency metric that surfaces strong performers even if they haven't sold the highest volume.



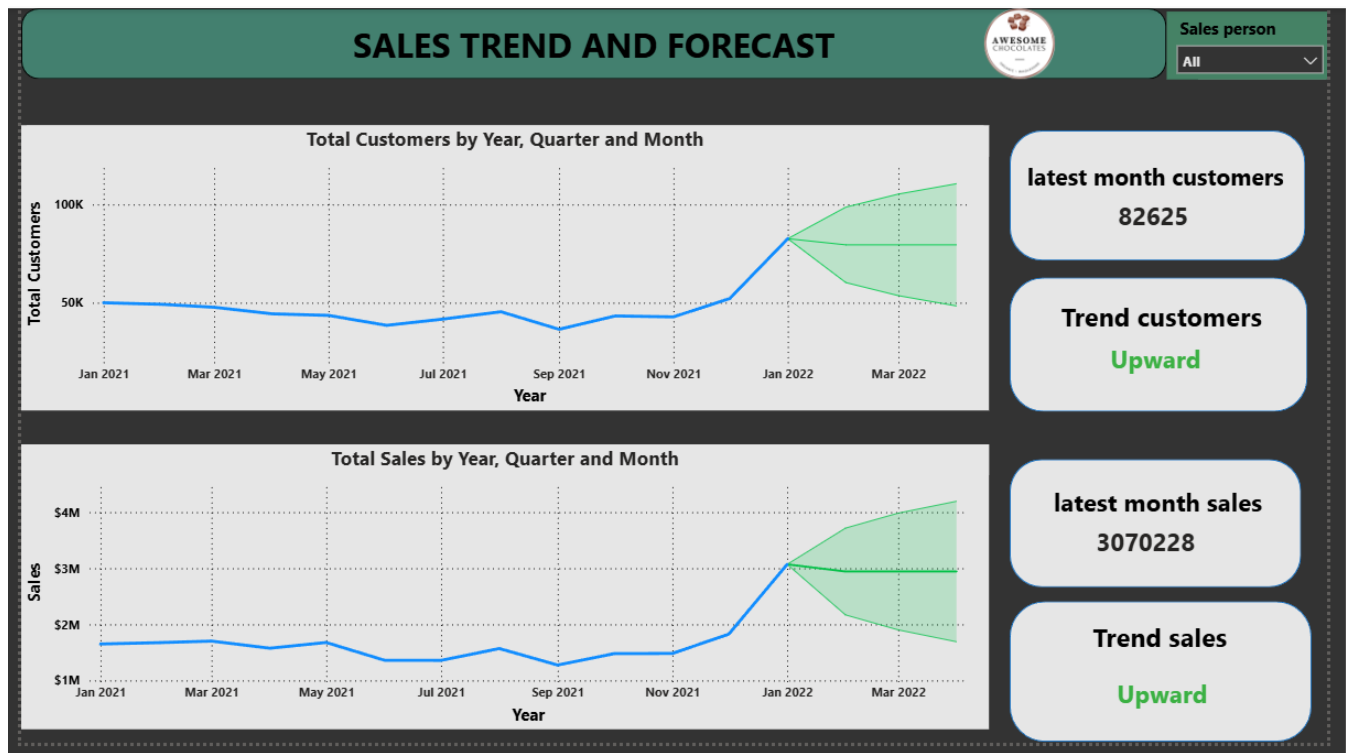
Page 2: Sales Performance Report

Scenario

A sales director wants to identify who to recognize this quarter. Sorting the leader board shows Van Tuxwell and Rafaelita Blaksland both earning \$17.0 per box the highest efficiency in the company which is a very different (and arguably more useful) signal than just looking at who sold the most boxes.

Page 3: Sales Trend and Forecast

This page answers: “Where is the business heading?” It tracks both customer count and sales revenue over time, with Power BI's native forecasting applied to project the next period with a 95% confidence interval, alongside simple Latest Month and Trend indicators.



Page 3: Sales Trend and Forecast

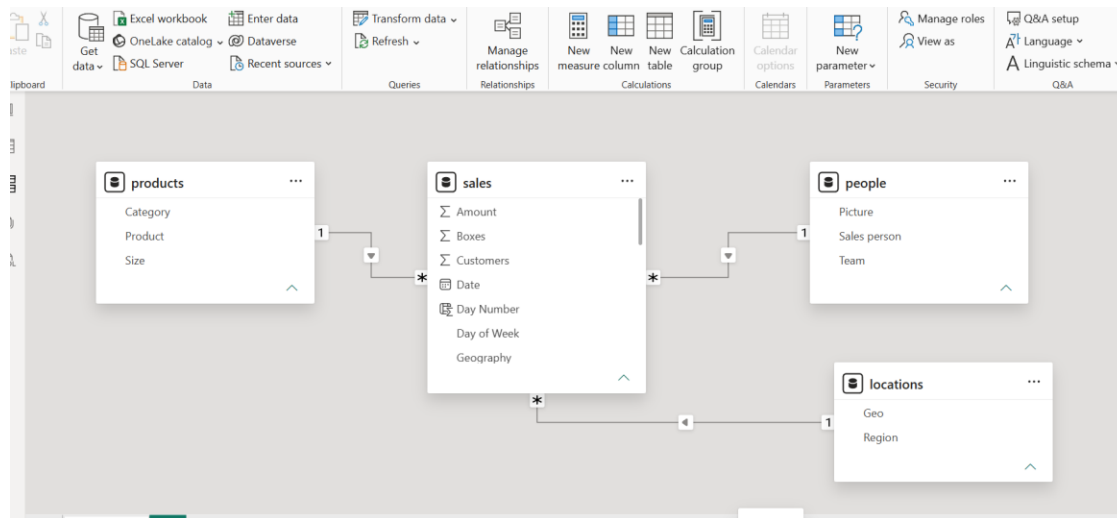
Scenario

Ahead of budget planning, a finance lead wants to know if current momentum is holding. The Trend card clearly reads “Upward” in green for both customers and sales, and the forecast band gives a defensible range to plan around rather than a single guessed number.

One decision I'm proud of on this page: I originally built a DAX-based numeric forecast card using a simple growth-rate formula. When I compared it against Power BI's native forecast, the two disagreed significantly my formula was reacting too strongly to one recent spike in the data. Rather than leave a misleading number on the dashboard, I removed that card and kept only the metrics I could stand behind the latest actual value, the trend direction, and Power BI's own confidence-interval forecast. I'd rather show less and have it be trustworthy.

The Data Model

Underneath all three pages is a simple star schema: one Sales fact table connected to Products, People, and Locations dimension tables. This structure keeps the model efficient and makes relationships between salespeople, geography, and product easy to filter across.



Star schema: Sales fact table with Products, People, and Locations dimensions

Summary of Solutions Addressed

- Replaced manual reporting with one easy to use dashboard that shows revenue, team performance, and sales by country all in one place.
- Looked beyond who sold the most, and also showed who sold the smartest using a "money per box" measure to give a fairer picture of performance.
- Found and fixed a hidden problem in the data (extra spaces in country names) that was causing some sales to be left out of the country charts by mistake.
- Built a short-term sales and customer forecast that gives a realistic range instead of one guessed number, so it's more trustworthy.
- Made all filters work together across every page, so you can pick one salesperson or team and see their full story without having to filter each page separately.

Growth Roadmap: Recommendations for the Company

Here's what I think Awesome Chocolates should focus on to grow sales, bring in more customers, and win in the right markets based on what the numbers actually show.

1. Boost Sales by Learning from Our Best People

Two of our salespeople, Van Tuxwell and Rafaelita Blaksland get more money for every box they sell than anyone else in the company. Before we just call that luck, we should find out what they're doing right maybe they sell better products, maybe they're great with customers, or maybe they just have a good sales technique. If we can teach that to others, it's an easy way to make more money without spending much.

We should also look at why some teams (Yummies and Delish) are outselling others (Jucies and Special). It might be that some teams have fewer people, cover tougher areas, or just need more coaching. Either way, it's worth digging into.

Right now, both our customer numbers and our sales are going up. This is a good time to add more sales staff or spend more on marketing it's easier to grow when things are already moving in the right direction.

2. Get More Customers

All six of our markets bring in roughly the same amount of money, somewhere between \$3.4 and \$3.8 million each. But that doesn't mean they all have the same number of customers. Some markets might be making that money from fewer people spending more, while others may have room to bring in more customers. We should track revenue per customer in each market to see where we can grow the customer base, not just sell more to the same people.

Since Van Tuxwell and Rafaelita Blaksland seem to be good with customers, we could also have them help bring in new customers, not just take care of existing ones.

Right now we only count how many customers we have, but we don't know who's new and who keeps coming back. If we started tracking that, we'd understand whether we're growing because of new customers or because old customers are buying more that changes what we should do next.

3. Which Markets to Focus On

New Zealand is currently our best market, bringing in \$3.78 million. We should figure out exactly what's working there so we can keep it going, not just watch it happen.

India, the UK, and Canada are close behind New Zealand only a few hundred thousand dollars apart. This tells us these markets aren't weaker, they just might need a bit more support, like more sales time or a small marketing push. We should try this in just one of these markets first and see if it works before doing it everywhere.

Australia and the USA are currently our weakest markets. Before spending more money there, we need to understand why is it because there just aren't many potential customers there, or are we simply not reaching the ones who are there? The answer changes what we should do.

4. Decisions to Make Right Now

- Decide if we want to set up a mentorship program where our best salespeople teach others what they do. This is the cheapest, easiest win available to us right now.
- Pick one mid-level market (India, UK, or Canada) and try investing a bit more there for one quarter. Compare the results before and after to see if it worked, instead of spreading a little money across all three at once.

What I Learned and Why It Matters

This project taught me more than I expected going in, largely because I kept running into real problems rather than a clean, guided tutorial path.

- **Data cleaning is not optional, and it's not obvious:** the whitespace bug taught me to always inspect a new dataset for hidden formatting issues before trusting any visual built on top of it. This is exactly the kind of habit that separates a reliable analyst from one who produces confident-looking but wrong numbers.
- **Not every model deserves to ship:** building the growth-rate forecast, comparing it against Power BI's own model, and then choosing to remove it when the numbers didn't hold up was the single most valuable moment in this project. It's easy to leave an impressive-looking number on a dashboard; it takes more discipline to check it against a second method and take it down when it doesn't survive scrutiny.
- **Context transition in DAX is a real skill, not a syntax trick:** debugging errors like “a single value for column cannot be determined” forced me to actually understand row context versus filter context, rather than just copying formulas that happened to work.
- **A dashboard is a conversation, not a report:** adding synced slicers and a dynamic “top market” insight card changed how the whole project felt from three separate static pages into one connected tool someone could actually explore.

Why this matters for where I'm headed: financial research analysis depends on exactly these habits questioning your own numbers, being transparent about the limits of your data, and choosing accuracy over a more impressive-looking result. This project was a low-stakes place to practice that discipline before I'm doing it with real financial data and real decisions riding on it.

Thank you for taking the time to look through this project. If you'd like to see the live dashboard file or discuss any part of the approach, I'd be glad to walk through it.

https://drive.google.com/file/d/1f5zo7iuvzBkuL9KJijRcrGYnONqfAG1k/view?usp=drive_link

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